- There are 5 Questions for 20 marks.
- Write your answers neatly, legibly and logically. Give supporting arguments for your answers. Keep your arguments thorough. Your arguments must follow from the basics discussed in class. You may use any result from class directly (without re-deriving).
- You do not need any calculators.
- Good luck!

1. [4] Each of the following statements presents an argument. For each, determine whether the conclusion logically follows from the premises, and *explain your reasoning*.
   (a) If an activity is deemed harmful to campus residents, it is banned on campus. Playing video games is banned on campus. Therefore, playing video games must be harmful.
   (b) If a student studies hard and attends all lectures, they will pass the exam. Ravi did not pass the exam. Therefore, Ravi did not study hard and Ravi did not attend all lectures.
   (c) A student does not get an A grade in EE9999 unless the student works hard or is very smart. Anita got an A grade in EE9999 and is very smart. Therefore, Anita must not have worked hard.
   (d) All dogs are mammals. Phani is a dog, so Phani is a mammal.

2. [4] Given $\underline{x} \in \mathbb{R}^4$, we define the difference operator ∇ as

$$\nabla \underline{x} = \begin{pmatrix} x_2 - x_1 \\ x_3 - x_2 \\ x_4 - x_3 \end{pmatrix} \text{ where } \underline{x} = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix}.$$

   Consider the function $f : \mathbb{R}^4 \mapsto \mathbb{R}^3$ defined as $f(\underline{x}) = \nabla \underline{x}$.
   (a) Is f one-one ?
   (b) Is f onto ?
   (c) Does f have a left inverse ? If so, find a left inverse of f.
   (d) Does f have a right inverse ? If so, find a right inverse of f.

3. [4] Suppose (x, y, z) lie on the surface of a sphere of radius r in $\mathbb{R}^3$. Then show that
   (a) $\max\{|x|, |y|, |z|\} \leq r$, and
   (b) $|x| + |y| + |z| \geq r$.
   (c) Which of (a), (b) above continue to hold for every choice of (x, y, z) inside the sphere ?
   (d) Is it possible to find (x, y, z) on the surface of sphere of radius r satisfying the following:
   x, y, z, r are natural numbers, $x = y$, and $r - z = 1$.

4. [4] Consider a group of n people. Some pairs of people in the group are friends (assume that friendship is mutual and that no one is friends with themselves). Consider the following statement (S): *If each person has exactly one friend, then n is even.*
   (a) What is the contraposition of S ?
   (b) What is the negation of S ?
   (c) Prove or disprove S.

5. [4] (a) Starting from sets S and T, define a set $f(S, T)$ as follows

$$a \in f(S,T) \iff ((a \in S \cup T) \wedge (a \in S \implies a \notin T))$$

   If $S = \{\text{apple}, \text{orange}, \text{grape}\}$, and $T = \{\text{grape}, \text{banana}\}$, what is $f(S, T)$?

(b) Starting from a set S, we define the set $P(S)$ as follows

$$A \in P(S) \iff (a \in A \implies a \in S).$$

If $S = \{\text{apple}, \text{orange}, \text{grape}\}$, what is $P(S)$?